

Where Linear Models Still Win

Supplementary reading, lesson 3 — Technologies for Artificial Intelligence

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Not examinable. It explains why the simplest model in the course is also the one you are most likely to meet in production.

Why a computer scientist should read this

Lesson 3 presented linear regression as the first method: exact solution, iterative solution, readable coefficients. It would be easy to file it under “the thing we did before the interesting models”.

That would be a mistake about the industry you are about to enter. Linear and logistic models remain in production, in some of the highest-stakes decisions made about people, decades after more accurate methods became available. The reasons are worth understanding, because they are not about accuracy — and because the day you propose replacing one, these are the objections you will meet.

The second half of this document is about **regression to the mean**, which is a statistical fact you have already met and a reasoning trap you have almost certainly fallen into.

1. Credit scoring: interpretability as a legal requirement

Consumer credit scoring is a large, mature, heavily regulated application, and much of it is logistic regression on a few dozen carefully engineered features — a “scorecard”.

That is not technological conservatism. Three forces hold it there.

Somebody has to be told why. A refused applicant is entitled to an explanation, and “the model said so” is not one. A scorecard produces a sentence: *your application scored below the threshold, mainly because of the number of recent credit enquiries*. A gradient-boosted ensemble can be probed with post-hoc tools, but the explanation is then an approximation of the model rather than the model itself — which is a materially weaker position in front of a regulator.

The model must be auditable years later. Scorecards are inspected, challenged and defended. A model whose behaviour can be read off a table of coefficients can be argued about; one that cannot has to be re-run.

Monitoring is easier. When a coefficient means “per standard deviation of this feature”, a shift in the input distribution has a readable consequence. The model tells you how it will misbehave before it does.

The accuracy gap is real but usually small — on tabular data with well engineered features, the difference between a good linear model and a good ensemble is often a couple of points. Whether those points are worth losing the three properties above is a business decision, and it frequently goes the other way.

2. Clinical risk scores: models a human must be able to apply

Medicine is full of scores that a doctor can compute at a bedside: a handful of inputs, small integer weights, a total that maps to a risk band.

These are almost always fitted as logistic regressions and then **deliberately degraded** — coefficients rounded to integers, continuous variables cut into bands. The rounding costs accuracy on purpose, in exchange for a model that can be applied without a computer, remembered, taught, and checked by a second clinician.

The engineering lesson generalises well beyond medicine: **the deployed model is the one that gets used**, and usability is part of performance. A model that requires infrastructure the setting does not have has an effective accuracy of zero there.

3. Insurance pricing: the model has to be filed

In much of Europe, insurance rating factors are filed with a regulator and must be justified — and some factors are prohibited outright. Generalised linear models dominate, partly because the structure makes the filing possible: each factor’s effect is a number that can be stated, defended, and constrained.

Note the shape of the constraint. It is not “the model must be accurate”. It is “the model must be *expressible*, and each part of it must be defensible on its own”. Flexible models are poor at that by construction: their power comes from interactions that resist being stated separately.

4. Econometrics: when the coefficient *is* the result

In economics and the social sciences, a regression is often run not to predict anything at all but to estimate one number — the effect of a policy, a price, a treatment — with an interval around it.

This is a different activity wearing similar clothes, and confusing the two causes real errors in both directions. A model with excellent predictive accuracy can have coefficients that mean nothing causally; a model with modest predictive accuracy can estimate one causal effect well. Lesson 3’s

warning about collinearity is the practical face of this: two columns carrying one fact predict fine together and neither coefficient can be interpreted alone.

5. Regression to the mean, and why the world seems to punish praise

The word “regression” comes from a phenomenon, not from the algorithm. Francis Galton, studying inherited height in the 1880s, observed that the children of unusually tall parents were on average taller than the population but *shorter than their parents* — “regression towards mediocrity”, in his phrase.

The mechanism is not biological. It is arithmetic, and it applies to anything measured with noise.

The picture. Any measurement is signal plus noise. An extreme measurement is extreme partly because the signal was high and partly because the noise happened to help. The signal persists on remeasurement; the luck does not. So the second measurement is, on average, less extreme.

That is all. And it produces one of the most robust illusions in human reasoning.

5.1 The illusion, in its classic form

Kahneman describes being told by flight instructors that praising a cadet for a good manoeuvre was followed by a worse one, while criticising a bad manoeuvre was followed by an improvement — and that they had therefore learned to criticise rather than praise.

Both observations were correct. The conclusion was wrong. An unusually good manoeuvre is followed by a worse one whatever the instructor says, and an unusually bad one by a better one, because performance varies and extremes are partly luck.

The instructors had run an uncontrolled experiment on themselves and drawn the opposite of the truth from it. **Reality had rewarded them for punishment and punished them for praise**, and nothing in the data would have told them otherwise.

5.2 Where you will meet it

- **Sports.** The “cover jinx” — an athlete featured after an exceptional run performs worse afterwards. Yes, and they would have anyway.
- **Medicine.** Patients enrol in a trial when their symptoms are at their worst. Many improve regardless of treatment, which is why an untreated control group exists.
- **Business.** The worst-performing region gets an intervention and improves. The intervention gets the credit.
- **Machine learning.** A feature selected because it scored highest in this sample will, on new data, score lower. Lesson 5 gives this its own name and measures it: the maximum of many noisy estimates is biased upward.

That last one is the connection worth carrying. Regression to the mean and the optimism of model selection are the same arithmetic seen from two directions, and recognising it in a story about flight instructors makes it much harder to miss in a leaderboard.

Where to look next

- Galton, “Regression Towards Mediocrity in Hereditary Stature” (1886) — short, readable, and the origin of the word.
- Kahneman, *Thinking, Fast and Slow*, the chapters on regression to the mean. The flight-instructor story is there in his own account.
- Rudin, “Stop Explaining Black Box Machine Learning Models for High Stakes Decisions and Use Interpretable Models Instead” (2019) — argues the position of this document considerably more forcefully than it does.
- Hastie, Tibshirani and Friedman, *The Elements of Statistical Learning*, chapter 3, for the statistical treatment of everything lesson 3 derived.